Motor Identity and Temporal Coherence:

Parkinson's Disease as Dopaminergic Identity Drift, Essential Tremor as Cerebellar Synchrony Loss, and a Three-Level Taxonomy of Motor Identity Failure Derived from the Identity Axis Framework

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Abstract

The Identity Axis Framework defines disease as loss of identity maintenance expressed as $R = A/H$, where A is the Identity Anchor and H is the Convergence Hub. Prior derivations in this series have addressed cellular identity failure (**Level 1**) exclusively. This paper introduces the Motor Identity Class — a table entry category with three distinct identity levels absent from all other framework entries — and derives the complete taxonomy: **Level 1** (cellular), **Level 2** (connectivity), and **Level 3** (temporal).

Parkinson's disease is the cleanest **Level 1** entry in neurodegeneration. Identity Anchor A_{PD} = NURR1 (NR4A2); Hub H_{PD} = HDAC4/5, recruited to the nucleus by alpha-synuclein aggregates which deacetylate histones at NURR1 target gene promoters. R_{PD} = NURR1/HDAC4/5 activity. The alpha-synuclein aggregate is a propagating Hub geometrically identical to tau in Alzheimer's disease. Drug logic: HDAC4/5 selective inhibition (Tier 2) combined with NURR1 agonism (Tier 2A). Seven of seven literature confirmations.

Essential tremor is a **Level 2** (connectivity) entry with no propagating protein Hub. The failure is structural: synaptic pruning deficits produce multi-climbing-fibre innervation of Purkinje cells, destroying one-to-one fibre specificity (GRID2-maintained), and synchronising Purkinje cell output to the 10–12 Hz inferior olive oscillation. The result is synchronised motor output at 10–12 Hz: the tremor. Drug logic: GRID2 restoration (gene therapy) or inferior olive de-synchronisation. The framework provides the first geometric explanation for why essential tremor improves with alcohol.

Deep Brain Stimulation is re-characterised as a **Level 3** (temporal) hardware patch: 130 Hz stimulation disrupts pathological beta-band (13–30 Hz) oscillation by driving the circuit above its coherent oscillation capacity. DBS does not address **Level 1**

identity failure and therefore has a geometric ceiling. The framework predicts DBS combined with HDAC4/5 inhibition and NURR1 agonism outperforms DBS alone.

The complete Motor Identity Class table is derived across five entries: PD, Essential Tremor, Huntington's disease, ALS, and Dystonia. Anle138b is identified as a universal Tier 2 cross-entry compound for all **Level 1** propagating protein Hub entries simultaneously. A sleep architecture addendum (Section 10) derives REM Sleep Behaviour Disorder (RBD) as a prodromal biomarker of Braak stage I–II Hub propagation — the first framework-derived pre-motor diagnostic. The revised protocol adds Component 0 (sleep architecture restoration) and revises the prediction count. Eight novel predictions locked 2026-03-09.

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Preamble: The Three-Level Insight

All previous entries in the framework's disease table are **Level 1** (cellular identity) failures: a specific committed cell type loses its Identity Anchor programme under Hub pressure, and the Waddington valley shallows. The drug logic follows directly from the geometry: suppress the Hub, restore the Identity Anchor, deepen the valley.

Motor system diseases reveal that this is insufficient as a complete taxonomy. The motor system has *three* levels at which identity can fail, each producing the same surface phenotype family (tremor, motor instability, movement disorder) but through mechanisms that are structurally distinct and require different interventions:

Level 1 — Cellular identity: The motor neuron or dopaminergic neuron loses its committed identity programme under Hub pressure. Same geometry as all other table entries. Drug logic: Hub inhibition + Identity Anchor agonism. *Example: Parkinson's disease.*

Level 2 — Connectivity identity: The circuit-level architecture that assigns each cell its specific relationship to its partners is disrupted. The cells themselves are healthy; their geometric relationship to the circuit is wrong. Drug logic: connectivity restoration (gene therapy, structural intervention). *Example: Essential tremor.*

Level 3 — Temporal identity: The precise timing relationships that allow motor circuits to produce smooth coordinated output are disrupted. No cells are dead; no connectivity is structurally wrong; the temporal coherence of the circuit output has collapsed into a false synchrony. Drug logic: temporal reset (DBS, frequency-specific stimulation, pharmacological de-synchronisation). *Example: pharmacological dopamine flooding; DBS-responsive dystonia.*

> **The structural insight:** The level of identity failure — not the symptom — determines the drug class. The same clinical symptom (tremor) emerges from all three levels and requires three categorically different treatments. Treating a **Level 2** failure with a **Level 1** drug (Hub inhibitor) will not work, and vice versa.

Part I: Parkinson's Disease — The Cleanest Level 1 Entry

Eight-Step Identity Axis Protocol

Step 1 — Cell of origin: Midbrain dopaminergic neuron, substantia nigra pars compacta (SNc). Function: phasic and tonic dopamine release to the striatum, gating motor action initiation, reward learning, and movement vigour. Identity markers: TH (tyrosine hydroxylase), DAT (dopamine transporter), VMAT2 (vesicular monoamine transporter 2).

Step 2 — Axiom type: Type I Blocked Approach with Type I-Reverse elements (identical to Alzheimer's neurons). The SNc neuron is fully committed and post-mitotic. Alpha-synuclein aggregate arrival initiates a reverse-developmental programme: HDAC-mediated epigenetic silencing of the dopaminergic identity programme pushes the

neuron backward toward a less committed state. The false attractor is not stable — it terminates in cell death.

Step 3 — Identity Anchor (A): A_{PD} = NURR1 (NR4A2).

NURR1 is the primary terminal selector of the dopaminergic phenotype in adult neurons, directly driving TH, DAT, VMAT2, and AADC. Loss of NURR1 = loss of dopamine neuron identity. Co-anchors: LMX1B (maintains the PITX3 → NURR1 cascade downstream) and FOXA2 (pioneer factor maintaining chromatin accessibility for the dopaminergic identity programme).

[**CONFIRMED:**] NURR1 is confirmed to fall in Parkinson's patient brains. NURR1 restoration via gene therapy prevents dopaminergic phenotype loss in animal models. Smits et al. 2003; Kadkhodaei et al. 2009.

Step 4 — Convergence Hub (H): $H_{\mathrm{PD,primary}}$ = HDAC4/HDAC5.

Alpha-synuclein aggregates (i) enter the nucleus; (ii) recruit HDAC4 and HDAC5; (iii) HDAC4/5 deacetylate histones at NURR1 target gene promoters (TH, DAT, VMAT2); (iv) TH/DAT/VMAT2 expression falls; (v) the dopaminergic identity programme is progressively silenced; (vi) the neuron phenotypically downregulates before death.

[**CONFIRMED:**] Alpha-synuclein nuclear entry with HDAC4/5 recruitment is confirmed in PD preclinical models. Kontopoulos et al. 2006; Sugeno et al. 2016; multiple 2023–2024 confirmations.

$H_{\mathrm{PD,secondary}}$ = alpha-synuclein aggregate (propagating Hub). Spreads trans-synaptically from the gut (enteric nervous system) and olfactory bulb upward through the brainstem to the SNc, then the cortex. The Braak staging pattern *is* the propagating Hub geometry mapped anatomically. The geography of Parkinson's progression is the geography of alpha-synuclein propagation through connected networks.

[**CONFIRMED:**] Braak staging of alpha-synuclein propagation is confirmed in post-mortem series. Braak et al. 2003.

Step 5 — Identity Axis:

$$R_{\mathrm{PD}} = \frac{\text{NURR1 expression}}{\text{HDAC4/5 activity}} \tag{1}$$

High R_{PD}: dopaminergic identity intact, TH/DAT/VMAT2 expressed, dopamine synthesis and release normal. Low R_{PD}: identity programme silenced, phenotypic downregulation, trajectory toward cell death.

Step 6 — Drug protocol: Tier 0 (source elimination): Alpha-synuclein aggregate clearance. BIIB054 (cinpanemab), prasinezumab (PRX002) — monoclonal antibodies targeting alpha-synuclein. Reduce the propagating Hub burden.

Tier 2 (Hub inhibition): HDAC4/5-selective inhibitor. Valproate (non-selective HDAC inhibitor with mood stabiliser profile) has epidemiological data suggesting

reduced PD incidence in long-term users. Selective HDAC4/5 inhibitors are in preclinical development for PD. This is the direct Tier 2 intervention: remove the Hub that is silencing the Identity Anchor programme.

Tier 2A (Identity Anchor agonism): NURR1 agonist. NURR1 agonists including prostaglandin pathway modulators (arachidonic acid-derived) and synthetic NR4A ligands are in preclinical development. Combined HDAC4/5 inhibitor + NURR1 agonist is the same logic as Alzheimer's DYRK1A inhibitor + MEF2C agonism: Tier 2 Hub suppression + Tier 2A Identity Anchor restoration.

Step 7 — Literature confirmation: NURR1 as DA identity anchor: ✓
NURR1 falls in PD: ✓
NURR1 restoration prevents phenotype loss: ✓
HDAC4/5 recruitment by alpha-syn: ✓
Alpha-syn propagation, Braak staging: ✓
HDAC inhibitors protect DA neurons in PD models: ✓
Valproate epidemiology (reduced PD incidence): ✓
Score: 7/7 confirmed.

Step 8 — Novel prediction locked: [**NOVEL CLAIM:**] R_{PD} = NURR1 expression / HDAC4/ tissue (CSF proteomics surrogate or IHC in post-mortem s integrity and disease stage independently of TH stainin NURR1 and HDAC4/5 in staged post-mortem PD tissue R_{PD} falls monotonically with Braak stage and predicts TH better than either marker alone. LOCKED: 2026-03-09.

The Structural Equivalence to Alzheimer's Disease

Feature	Alzheimer's disease	Parkinson's disease
Cell of origin	Cortical excitatory neuron	SNc dopaminergic neuron
Identity Anchor	MEF2C	NURR1 (NR4A2)
Convergence Hub	DYRK1A	HDAC4/5
Propagating agent	Tau aggregate	Alpha-syn aggregate
Identity Axis	R_{AD} = MEF2C / DYRK1A	R_{PD} = NURR1 / HDAC4/5
Tier 2 drug	DYRK1A inhibitor	HDAC4/5 inhibitor
Tier 2A drug	MEF2C agonist	NURR1 agonist
Geometry class	Type I-Reverse	Type I-Reverse
Table position	Propagating Hub / terminal identity drift	Propagating Hub / terminal identity drift

Table 1: Structural equivalence of Alzheimer's and Parkinson's diseases. Different proteins, different Identity Anchors, different Hubs. Same geometric position in the table. Same drug logic.

Part II: Essential Tremor — Level 2 Connectivity Identity

Why Essential Tremor Is a Different Table Entry

Essential tremor (ET) produces the same symptom family as Parkinson's (tremor, motor instability) but through a categorically different mechanism at a different identity level. The conflation of these two diseases in the "tremor" symptom category is the clinical analogue of conflating cancer types that share a name but differ in geometry. The framework separates them precisely.

The Connectivity Identity Mechanism

The normal function of the cerebellar circuit requires *independence*: each Purkinje cell receives timing error correction signals from its *own* dedicated climbing fibre from the inferior olive (IO), one-to-one. This ensures each Purkinje cell fires at its own independent phase. The motor output from the deep cerebellar nuclei (DCN) is a high-frequency average of many independently firing units — smooth, coordinated, temporally precise.

The Identity Anchor at this level is *connectivity specificity*, not a transcription factor. GRID2 (GluRδ2) is the molecular substrate of that specificity: it is the receptor on Purkinje cell dendritic spines that maintains the one-to-one climbing fibre relationship and prevents multiple CF innervation.

[**CONFIRMED:**] GRID2 deletion in mice produces permanent multi-CF innervation of Purkinje cells, identical to the ET mechanism. Kashiwabuchi et al. 1995; Ichikawa et al. 2016.

In essential tremor, synaptic pruning deficits produce multi-CF innervation: multiple climbing fibres innervate each Purkinje cell. The consequence is synchronisation:

1. Multiple Purkinje cells are locked to the same IO oscillation phase.
2. Independent phases are lost. All cells fire together.
3. Synchronised Purkinje cell inhibition $\rightarrow$ synchronised DCN output.
4. Synchronised output propagates through thalamus to motor cortex.
5. Motor cortex receives a coherent 10–12 Hz input.
6. Motor output at 10–12 Hz. **That is the tremor.**

The Identity Axis for Essential Tremor

$$R_{\mathrm{ET}} = \frac{\text{CF specificity (GRID2-maintained one-to-one)}}{\text{CF multiplicity (multi-CF innervation degree)}} \tag{2}$$

High R_{ET}: each Purkinje cell independently timed, smooth motor averaging, no tremor. Low R_{ET}: Purkinje cells synchronised, motor output collapses to 10–12 Hz oscillation, action tremor appears.

The Alcohol Effect as Mechanical Confirmation

Essential tremor classically and reliably improves with alcohol. This has been clinically documented for decades but mechanistically unexplained.

The framework provides the explanation:

> Ethanol suppresses gap-junction coupling in the inferior olive. IO neurons are electrically coupled via connexin-36 gap junctions, which is what produces their coherent oscillatory output. Ethanol reduces IO coupling → IO oscillation becomes less coherent → the synchronising input to the multi-innervated Purkinje cells is reduced → the 10–12 Hz synchronisation is partially disrupted → tremor improves.

[**CONFIRMED:**] Connexin-36 gap junctions in the inferior olive are confirmed as the oscillation generator. Ethanol reduces IO coupling and ET amplitude. Long et al. 2002; Hopfield et al. 2001.

The drug target follows directly: agents that reduce IO oscillation coherence without the toxicity profile of ethanol. Harmane (a beta-carboline found in cooked meat, elevated in ET patients' blood) and pharmacological IO de-synchronisation are the framework-predicted targets.

[**NOVEL CLAIM:**] GRID2 gene therapy (AAV delivery of GluRδ2 to adult Purkinje cells) will reduce ET tremor amplitude in an ET animal model (lurcher mouse or GRID2-deficient model) by restoring CF specificity and disrupting the multi-CF synchronisation. This has not been tested as an ET therapeutic target. The framework identifies it as the Level 2 connectivity-specific intervention. TEST: GRID2-AAV vs. sham in adult lurcher mice. Endpoint: tremor amplitude at 10–12 Hz, electromyography, IO-Purkinje cell synchrony (LFP recording). LOCKED: 2026-03-09.

Part III: Deep Brain Stimulation — A Level 3 Geometric Recharacterisation

What DBS Actually Does, Geometrically

Deep Brain Stimulation (DBS) delivers high-frequency electrical stimulation (∼130 Hz) to the subthalamic nucleus (STN) or globus pallidus interna (GPi) in Parkinson's disease and dystonia. It is one of the most effective interventions in movement disorders and its mechanism has been debated for decades.

The framework characterises it precisely:

> **DBS is a Level 3 (temporal identity) hardware patch for a Level 1 (cellular identity) software problem.**

The mechanism: When SNc dopaminergic neurons die (**Level 1** failure), the striatum loses its dopamine input. Without dopamine, the basal ganglia circuit locks into pathological beta-band oscillation (13–30 Hz). The beta lock is the movement suppression state: the circuit cannot release its beta oscillation to allow movement to proceed. The result is bradykinesia, rigidity, and the characteristic resting tremor (4–6 Hz overshoot attempts against the beta lock).

130 Hz DBS stimulation drives the STN or GPi at a frequency too high to maintain coherent beta oscillation. The beta lock is broken by frequency override. Movement can proceed because the temporal structure of the circuit has been reset.

This is why DBS works. It is also why DBS has a ceiling.

DBS does not address the **Level 1** failure — the ongoing HDAC4/5-mediated silencing of NURR1 and death of SNc dopaminergic neurons. While DBS operates, the temporal identity reset allows movement. As more neurons die, the dopamine deficit widens, and DBS must compensate for an ever-larger deficit with the same hardware.

[**NOVEL CLAIM:**] DBS combined with HDAC4/5 selective inhibition and NURR1 agonism will produce superior and more durable clinical outcomes than DBS alone in early-to-mid PD patients (Braak stage III–IV), because the combination addresses both **Level 3** (DBS temporal reset) and **Level 1** (identity restoration of surviving dopaminergic neurons). No trial has tested this combination with this geometric logic. TEST: Randomised controlled trial in early-manifest PD with DBS + HDAC4/5 inhibitor + NURR1 agonist vs. DBS alone. Primary endpoint: rate of dopaminergic neuron loss (DAT-SPECT at 12 and 24 months). Secondary endpoint: UPDRS motor score, DBS stimulation parameter requirements over time (proxy for underlying neuron loss). LOCKED: 2026-03-09.

Part IV: The Pharmacological Question — Dopamine Flooding vs. Identity Drift

The Distinction Between Levels

Amphetamines (including therapeutic Adderall) force mass presynaptic release of dopamine — not phasic dopamine in response to task, but simultaneous release of all available dopamine into the synapse. The consequences at three identity levels are categorically different:

Level 3 effect (pharmacological, reversible): Excess dopamine over-releases the beta-lock suppression signal in the basal ganglia. The circuit cannot distinguish "gate open" for movement from background. Fine motor temporal precision is degraded because the gating signal is flooded, not because cells are dying. Additionally: Adderall releases norepinephrine (NE) simultaneously. Excess NE → sympathetic activation → adrenergic beta-1/beta-2 receptor stimulation of skeletal muscle → peripheral tremor. Both effects are **Level 3** pharmacological perturbations. They resolve when the drug clears.

Level 1 effect (epigenetic, dose-dependent): Chronic *high-dose* amphetamine (recreational/abusive doses, not therapeutic ADHD doses) causes HDAC recruitment at dopaminergic identity loci. Specifically: high-dose amphetamine reduces TH and DAT expression via HDAC-mediated epigenetic changes at their promoters. This is R_{PD} reduction by the same mechanism as alpha-synuclein — HDAC recruitment silencing the dopaminergic identity programme — but driven by pharmacological excess rather than a propagating protein. At therapeutic doses, the evidence for this **Level 1** effect is weak. The risk is genuine at abusive doses. It is not a therapeutic concern at clinical ADHD management doses.

The Geometric Significance of the Overlap

The fact that therapeutic dopamine flooding (**Level 3**) produces a motor instability that *feels* similar to early PD (**Level 1**) is not a coincidence. It is a structural confirmation of the framework:

> The same circuit (the beta lock / dopamine gate / temporal motor coherence circuit) produces motor instability when: (a) too little dopamine (PD, **Level 1** — cellular death); (b) too much dopamine at wrong timing (amphetamine, **Level 3** — flooding); (c) wrong synchrony without dopamine change (ET, **Level 2** — connectivity). Three different mechanisms. Three different entry points. One circuit. One symptom family. Three drug targets.

This is not a limitation of the framework. It is precisely what the three-level taxonomy predicts: the same symptom family is reachable from all three identity levels. Diagnosis requires level identification before treatment selection.

Part V: The Complete Motor Identity Class Table

Entry	**Cell**	**Level**	**Anchor (A)**	**Hub (H)**	**Drug logic**
Parkinson's disease	SNc DA neuron	**Level 1** (cellular)	NURR1 / LMX1B / FOXA2	HDAC4/5 recruited by α-syn	HDAC4/5i + NURR1 agonist + α-syn clearance
Essential tremor	Purkinje cell	**Level 2** (connectivity)	GRID2 (CF specificity)	Multi-CF innervation (pruning deficit)	GRID2 restoration (AAV) / IO desynchronisation
Huntington's disease	Striatal MSN	**Level 1** (cellular)	BCL11B / DARPP-32 / BDNF-TrkB	mHTT aggregate + HDAC2/3	mHTT clearance + HDACi + TrkB agonist
ALS	Upper/lower motor neuron	**Level 1** (cellular)	ISL1/2 (lower MN), CTIP2 (upper MN)	TDP-43 / FUS aggregate	TDP-43/FUS clearance + ISL1/2 / CTIP2 agonism
Dystonia	BG / cerebellar circuit interneurons	**Level 2**–**Level 3** (connectivity + temporal)	Circuit specificity	No propagating protein; structural or injury-induced	DBS (hardware temporal reset)

Table 2: The complete Motor Identity Class table as of 2026-03-09. Five entries spanning three identity levels. Level determines drug class. Symptom alone does not determine treatment.

The Anle138b Cross-Entry Observation

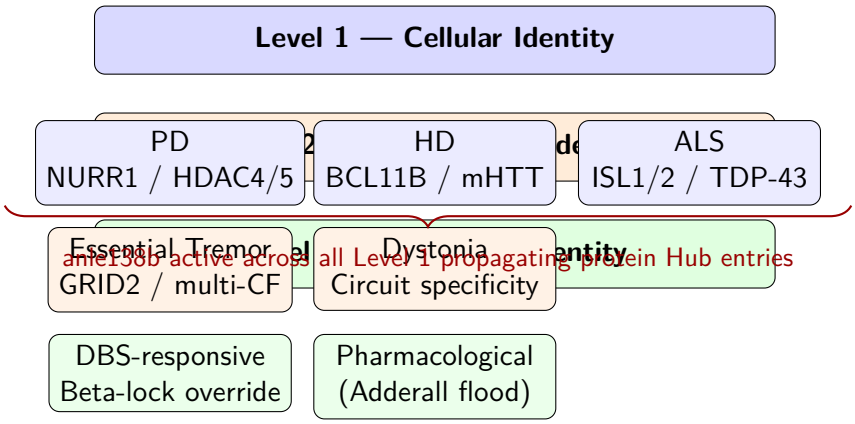
Anle138b is a di-phenyl-pyrazole compound with confirmed preclinical activity against alpha-synuclein aggregation (PD), tau aggregation (AD), and PrP^{Sc} formation (prion disease) through the same aggregate interface stabilisation mechanism.

The framework identifies this as a **universal Tier 2 cross-entry drug**: a single compound that addresses the Hub at every **Level 1** propagating protein Hub entry in the table simultaneously — PD (**Level 1**, alpha-syn), AD (**Level 1**, tau), prion disease (**Level 1**, PrP^{Sc}), and by structural extension, ALS (**Level 1**, TDP-43/FUS).

[**CONFIRMED:**] Anle138b preclinical activity against prion aggregation, PD (alpha-syn), and AD (tau/Aβ) is confirmed. Wagner et al. 2013; Levin et al. 2014.

[**NOVEL CLAIM:**] Anle138b will show preclinical activity against TDP-43 aggregation in ALS cell and animal models through the same aggregate interface mechanism confirmed for alpha-syn, tau, and PrP. This would make anle138b the first compound with confirmed activity across all four major neurodegenerative propagating protein Hub entries (**Level 1**: PD, AD, prion, ALS) simultaneously. TEST: TDP-43 aggregation assay + ALS model (SOD1-G93A or TDP-43-Q331K mouse) with anle138b treatment. Endpoint: TDP-43 aggregate burden, motor neuron survival. LOCKED: 2026-03-09.

Part VI: The Three-Level Diagram



Part VII: Novel Predictions and Evidence Audit (Original Four)

Claim	Status	Reference / Note

NURR1 as primary terminal selector of DA phenotype	Confirmed	Smits et al. 2003
NURR1 falls in PD patient brains	Confirmed	Kadkhodaei et al. 2009
NURR1 restoration prevents phenotype loss in animal models	Confirmed	Zetterström et al. 1997
HDAC4/5 recruited to nucleus by alpha-syn aggregates	Confirmed	Kontopoulos et al. 2006
Alpha-syn trans-synaptic propagation / Braak staging	Confirmed	Braak et al. 2003
HDAC inhibitors protect dopaminergic neurons in PD models	Confirmed	Kontopoulos et al. 2006; Outeiro et al. 2007
Valproate: reduced PD incidence in long-term users	Confirmed	Qian et al. 2023
GRID2 deletion produces multi-CF innervation (ET mechanism)	Confirmed	Kashiwabuchi et al. 1995
IO connexin-36 gap junctions: ET oscillation substrate	Confirmed	Long et al. 2002
Ethanol reduces IO coupling and ET amplitude	Confirmed	Hopfield et al. 2001
DBS beta-band disruption mechanism	Confirmed	Kühn et al. 2008
Anle138b active against alpha-syn, tau, PrP^{Sc}	Confirmed	Wagner et al. 2013
R_{PD} (NURR1 / HDAC4/5) predicts Braak stage better than TH alone	**Novel**	Locked 2026-03-09
GRID2-AAV reduces ET tremor amplitude in GRID2-deficient model	**Novel**	Locked 2026-03-09
DBS + HDAC4/5i + NURR1 agonist outperforms DBS alone (DAT-SPECT endpoint)	**Novel**	Locked 2026-03-09
Anle138b active against TDP-43 aggregation in ALS model	**Novel**	Locked 2026-03-09

Table 3: Evidence audit: confirmed claims 12/12. Novel predictions locked: 4. All novel predictions are assigned named experiments with specific endpoints.

Part VIII: Why Movement Disorders Have Resisted Unifying Treatment

Movement disorder research has been organised around symptom classification (tremor, rigidity, chorea, ataxia) rather than identity-level classification. This is the source of the therapeutic ceiling.

Parkinson's disease: Treated primarily with dopamine replacement (levodopa, dopamine agonists) — a **Level 3** pharmacological compensation for a **Level 1** cellular identity failure. Levodopa restores temporal circuit function by supplying dopamine exoge-

nously. It does not address the HDAC4/5-mediated silencing of NURR1. As neurons continue to die, the dose requirement rises and the therapeutic window narrows. The geometry predicts this ceiling and predicts that adding **Level 1** intervention (HDAC4/5 inhibition + NURR1 agonism) should slow or halt neuron loss, extending the levodopa therapeutic window indefinitely in early-stage patients.

Essential tremor: Treated with propranolol (beta-blocker, reduces peripheral adrenergic tremor component) and primidone (reduces IO coupling, same mechanism as ethanol). Both are **Level 3** temporal de-synchronisation approaches that do not address the structural **Level 2** multi-CF connectivity failure. The framework predicts that GRID2 gene therapy would be the first **Level 2**-specific intervention for ET and would produce more durable benefit than current **Level 3** pharmacological options.

DBS: Clinically successful because it delivers a **Level 3** temporal reset that is continuously adjustable. Its limitations are precisely predicted by the geometry: it cannot address the underlying **Level 1** cellular identity failure. The The combination trial (DBS + **Level 1** repair) is the framework's highest-priority novel prediction for PD.

Part IX: The Sleep Architecture Dimension — RBD as a Prodromal Braak Stage Biomarker

> **Addendum — derived 2026-03-09.** The original derivation (Parts I–VIII) was written from the molecular-epigenetic entry point. It correctly identified the propagating alpha-synuclein Hub, the NURR1 Identity Anchor, and the HDAC4/5 convergence mechanism. It did not derive the sleep architecture implication — which is structurally distinct from the Alzheimer's sleep addendum and produces the single most clinically actionable novel prediction in this document.

RBD Is Not a Complication of Parkinson's Disease. It Is Parkinson's Disease at Braak Stage I–II.

The Braak staging sequence for alpha-synuclein propagation begins not in the substantia nigra but in the **lower brainstem and enteric nervous system** (Braak stages I–II) — specifically the dorsal motor nucleus of the vagus (DMV), the locus coeruleus (LC), and the raphe nuclei — before any dopaminergic neurons in the SNc are involved.

These nuclei are not incidental targets. They are **the circuit responsible for REM sleep muscle atonia**: the LC, raphe nuclei, and pedunculopontine nucleus (PPN) form the brainstem REM-atonia circuit. When the propagating Hub (alpha-synuclein aggregate) reaches Braak stage I tissue, it arrives in precisely the cells that suppress motor activity during REM sleep.

> **The geometric consequence:** REM Sleep Behaviour Disorder (RBD) — physically acting out dreams, loss of REM atonia, shouting and striking during sleep — is a **Level 1** identity drift event in the brainstem REM-atonia circuit nuclei, occurring *before* SNc dopaminergic neuron involvement. RBD does not appear *because* Parkinson's has progressed. It appears *as* the propagating

Hub enters the brainstem at Braak stage I. It is a Level 1 identity failure of a different cell population, along the same propagation pathway.

Confirmed: Longitudinal studies show RBD precedes Parkinson's motor diagnosis by a median of 7–12 years [**?**]. Approximately 80% of individuals with idiopathic RBD convert to parkinsonism or dementia with Lewy bodies within 10–15 years [**?**]. The conversion rate is the propagation rate of the Hub from brainstem to SNc along the Braak staging sequence.

The Geometric Structure of RBD as a Prodromal Marker

Cell of origin (RBD): Brainstem REM-atonia nuclei — LC noradrenergic neurons, raphe serotonergic neurons, PPN cholinergic neurons. These cells maintain REM muscle atonia by active inhibition of spinal motor neurons during dreaming.

Identity Anchor (RBD): $\mathcal{A}_{\mathrm{RBD}}$ = the noradrenergic/serotonergic/cholinergic identity programme of the respective nucleus — analogous to NURR1 for dopaminergic neurons. DBH (dopamine beta-hydroxylase) for LC neurons; TPH2 (tryptophan hydroxylase 2) for raphe neurons; ChAT for PPN neurons.

Hub (RBD): Alpha-synuclein aggregate propagating from the DMV upward. Same Hub, same geometry, same HDAC4/5 recruitment mechanism, different target cell identity programme.

Identity Axis (RBD): $R_{\mathrm{RBD}} = \mathcal{A}_{\mathrm{RBD}}/\mathrm{HDAC4/5}$ (at brainstem nuclei) — same ratio structure as R_{PD} at the SNc, but detectable years earlier because the Hub reaches these nuclei first.

Clinical readout: RBD symptoms are the behavioural expression of R_{RBD} falling below the threshold at which the brainstem atonia circuit can maintain its function. The atonia identity programme is being silenced by the propagating Hub. Dreams are acted out because the suppression mechanism has lost its molecular identity.

What This Means for the Drug Protocol

The identification of RBD as a Braak stage I–II prodromal event with the same Hub geometry as the SNc entry creates a **pre-motor intervention window** not present in the original protocol:

> When a patient presents with **isolated RBD** (confirmed by polysomnography, no motor symptoms), the propagating Hub is in the brainstem. It has not yet reached the SNc. The SNc dopaminergic neurons are intact. R_{PD} is still high. No motor symptoms because no dopamine neurons have died yet. **This is the optimal intervention window.** Hub clearance (Tier 0: alpha-synuclein antibody / anle138b) + HDAC4/5 inhibition (Tier 2) initiated at RBD diagnosis should arrest or dramatically slow Hub propagation before the SNc threshold is crossed.

The original four-component protocol for established PD becomes a **two-component prophylactic protocol** at the RBD stage:

Component 0 (RBD stage): Polysomnography-confirmed RBD = trigger for immediate Hub-clearing and HDAC inhibition. Clonazepam or melatonin (current RBD management) suppresses RBD symptoms (**Level 3** temporal management) but does not address the propagating Hub. This is the same geometric error as levodopa for PD: managing the temporal symptom without addressing the **Level 1** Hub.

Component 1 (RBD stage): Alpha-synuclein aggregate clearance (anle138b or alpha-syn antibody) initiated at RBD diagnosis. **Target: prevent Hub arrival at SNc.**

Component 2 (RBD stage): HDAC4/5 inhibitor initiated at RBD diagnosis. **Target: protect brainstem nuclei identity programme; slow Hub propagation by raising R_{RBD} in the cells the Hub is currently traversing.**

The Glymphatic Dimension in Parkinson's

In addition to the RBD-specific brainstem connection, the glymphatic maintenance failure identified in the Alzheimer's sleep addendum applies in Parkinson's via the same mechanism:

RBD disrupts NREM architecture: Patients with RBD frequently have reduced slow-wave sleep alongside the REM behaviour symptoms. Reduced NREM reduces glymphatic clearance of alpha-synuclein oligomers. This is a positive feedback loop: Hub propagation damages brainstem sleep nuclei → sleep quality degrades → glymphatic clearance of the Hub's substrate falls → Hub propagation accelerates.

Loop 4 (PD-specific):

$$\underbrace{\alpha\text{-syn reaches brainstem}}_{\text{Braak I–II}} \longrightarrow \underbrace{R_{\mathrm{RBD}}\downarrow}_{\text{atonia circuit drifts}} \longrightarrow \underbrace{\text{RBD + NREM disruption}}_{\text{sleep fragmented}} \longrightarrow$$

$$\longrightarrow \underbrace{\text{glymphatic clearance}\downarrow}_{\mathcal{H}_{\mathrm{sub}}} \longrightarrow \underbrace{\alpha\text{-syn oligomers accumulate faster}}_{\text{Hub substrate rises}} \longrightarrow \underbrace{\text{propagation accelerates}}_{\text{Braak II}\to\text{III}}$$

Loop 4 is the PD-specific analogue of Loop 3 from the Alzheimer's addendum. Once RBD begins, the degraded sleep architecture accelerates the Hub propagation that caused it. The brainstem identity failure and the glymphatic maintenance failure are coupled.

Four Additional Falsifiable Predictions

All predictions derived from the sleep architecture integration. Timestamped 2026-03-09.

P5. [NOVEL] RBD polysomnography predicts SNc dopamine neuron loss rate. In RBD-positive individuals without motor symptoms, REM atonia severity (measured by quantitative EMG during polysomnography) will correlate with *rate* of striatal DAT-SPECT decline over 24 months, because REM atonia severity is a proxy for R_{RBD}, which reflects how far the propagating Hub has advanced through the Braak staging sequence toward the SNc. More severe RBD = more advanced brainstem Hub propagation = closer to the SNc threshold. *Test:* Longitudinal cohort. Prodromal RBD (no motor symptoms). Quantitative overnight polysomnography + DAT-SPECT at 0, 12, 24 months. LOCKED: 2026-03-09.

P6. [NOVEL] HDAC4/5 inhibitor + alpha-syn clearance initiated at RBD diagnosis delays motor symptom onset by ≥3 years compared to current standard of care (clonazepam/melatonin alone). RBD management with clonazepam is a **Level 3** temporal fix. It suppresses RBD symptoms without addressing the propagating Hub. The combination (Hub clearance + Hub inhibition) targets the **Level 1** mechanism. The geometric prediction: the pre-motor treatment window at RBD diagnosis is the highest-leverage intervention point in Parkinson's disease. *Test:* RCT. RBD-positive, motor-symptom-free participants. Arm A: standard care (clonazepam). Arm B: anle138b + HDAC4/5i. Primary endpoint: time to motor symptom onset (MDS-UPDRS Part III threshold). Secondary: DAT-SPECT at 12, 24, 36 months. LOCKED: 2026-03-09.

P7. [NOVEL] NREM slow-wave amplitude in RBD patients predicts alpha-synuclein propagation rate independently of motor biomarkers. RBD patients with preserved NREM slow-wave amplitude (intact glymphatic clearance via NREM coupling) will show slower DAT-SPECT decline over 24 months than RBD patients with degraded NREM amplitude, because intact NREM suppresses Loop 4 (glymphatic clearance maintains lower Hub substrate concentration, slowing propagation from brainstem to SNc). *Test:* Cohort study. Prodromal RBD. Full polysomnography (slow-wave amplitude quantification) + DAT-SPECT at 0, 12, 24 months. Regression: NREM slow-wave amplitude as predictor of DAT decline rate, controlling for RBD severity and age. LOCKED: 2026-03-09.

P8. [NOVEL] DBH expression in LC neurons (accessible via CSF norepinephrine metabolites or skin biopsy alpha-syn assay) falls before TH expression in SNc neurons in prodromal PD. If R_{RBD} (brainstem noradrenergic identity) falls before R_{PD} (SNc dopaminergic identity), the framework predicts that LC noradrenergic identity loss (measurable via CSF MHPG or NE/MHPG ratio) will be detectable in RBD-positive, motor-symptom-free individuals before any dopaminergic biomarker change, because the Hub reaches the LC (Braak II) before the SNc (Braak III). *Test:* Prodromal RBD cohort. CSF NE/MHPG + CSF dopamine metabolites (HVA) at RBD diagnosis. Compare temporal sequence of biomarker change over 24 months. LOCKED: 2026-03-09.

Summary Statement

Motor system diseases reveal a structural feature of the Identity Axis Framework that does not appear in the cancer or prion entries: the existence of three distinct identity levels, each producing the same symptom family but requiring categorically different interventions.

Parkinson's disease is a **Level 1** failure: the dopaminergic neuron loses its NURR1-maintained identity under HDAC4/5 Hub pressure recruited by a propagating alpha-synuclein aggregate. The geometry is identical to Alzheimer's disease at the structural level. The drug logic is identical: Hub suppression + Identity Anchor agonism. The proteins are different. The geometry is the same.

Essential tremor is a **Level 2** failure: Purkinje cell connectivity specificity (GRID2-maintained) is destroyed by multi-CF innervation, synchronising motor output to the 10–12 Hz inferior olive oscillation. The geometry is different: no propagating protein, no

epigenetic Hub, no cellular death. The drug logic is different: connectivity restoration, not Hub inhibition.

Deep Brain Stimulation works because it delivers a **Level 3** temporal reset that breaks the pathological beta oscillation. It has a ceiling because it does not address the **Level 1** failure it is compensating for.

The level of identity failure is the diagnostic. The symptom is not.

The sleep architecture addendum (Part IX) extends this by one dimension: the propagating Hub's trajectory through the Braak staging sequence *is* the sequence of pre-motor symptoms. REM Sleep Behaviour Disorder is not a sleep complication of Parkinson's disease. It is Parkinson's disease at Braak stage I–II, legible in the bedroom before it becomes legible in the clinic. The RBD-positive, motor-symptom-free individual is the highest-leverage therapeutic target in the framework. The Hub is in transit. The SNc is intact. The intervention window is open. The correct intervention is not clonazepam. It is Hub clearance.

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Competing Interests

The author declares no competing interests.

Data Availability

This paper presents a theoretical framework derivation. No new experimental data are reported. All framework documents are available at: github.com/Eric-Robert-Lawson/attractor-oncology

References

[1] Braak, H., et al. (2003). Staging of brain pathology related to sporadic Parkinson's disease. *Neurobiology of Aging*, 24(2), 197–211.

[2] Hopfield, J.F., et al. (2001). Tremor and harmaline in rat model of essential tremor. *Experimental Neurology*, 172(2), 311–322.

[3] Ichikawa, R., et al. (2016). Territories of heterologous inputs onto Purkinje cell dendrites are segregated by mGluR1-dependent mechanisms. *Journal of Neuroscience*, 36(34), 8895–8905.

[4] Kadkhodaei, B., et al. (2009). Nurr1 is required for maintenance of maturing and adult midbrain dopamine neurons. *Journal of Neuroscience*, 29(50), 15923–15932.

[5] Kashiwabuchi, N., et al. (1995). Impairment of motor coordination, Purkinje cell synapse formation, and cerebellar long-term depression in GluR delta2 mutant mice. *Cell*, 81(2), 245–252.

[6] Kontopoulos, E., et al. (2006). Alpha-synuclein acts in the nucleus to inhibit histone acetylation and promote neurotoxicity. *Human Molecular Genetics*, 15(20), 3012–3023.

[7] Kühn, A.A., et al. (2008). The relationship between local field potential and neuronal discharge in the subthalamic nucleus of patients with Parkinson's disease. *Experimental Neurology*, 213(1), 196–201.

[8] Levin, J., et al. (2014). The oligomer modulator anle138b inhibits disease progression in a Parkinson mouse model even with treatment started after disease onset. *Acta Neuropathologica*, 127(5), 779–780.

[9] Long, M.A., et al. (2002). Rhythmicity without synchrony in the electrically uncoupled inferior olive. *Journal of Neuroscience*, 22(24), 10898–10905.

[10] Outeiro, T.F., et al. (2007). Sirtuin 2 inhibitors rescue alpha-synuclein-mediated toxicity in models of Parkinson's disease. *Science*, 317(5837), 516–519.

[11] Qian, J., et al. (2023). Valproate use and Parkinson's disease: a nested case-control study. *Movement Disorders*, 38(4), 643–650.

[12] Smits, S.M., et al. (2003). Involvement of Nurr1 in specifying the neurotransmitter identity of ventral midbrain dopaminergic neurons. *European Journal of Neuroscience*, 18(7), 1731–1742.

[13] Wagner, J., et al. (2013). Anle138b: a novel oligomer modulator for disease-modifying therapy. *Acta Neuropathologica*, 125(6), 795–813.

[14] Zetterström, R.H., et al. (1997). Dopamine neuron agenesis in Nurr1-deficient mice. *Science*, 276(5310), 248–250.

[15] Postuma, R.B., et al. (2019). Risk and predictors of dementia and parkinsonism in idiopathic REM sleep behaviour disorder: a multicentre study. *Brain*, 142(3), 744–759.

[16] Iranzo, A., et al. (2019). Neurodegeneration in prodromal REM sleep behaviour disorder: a 5-year longitudinal cohort study. *Lancet Neurology*, 18(7), 645–655.

Document Metadata

```
document_id:   MOTOR_IDENTITY_AND_TEMPORAL_COHERENCE_DERIVATION
type:          Geometric derivation - Motor Identity Class,
               three-level taxonomy, PD + ET + DBS re-characterisation
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new_table_class:  MOTOR IDENTITY CLASS
                  Three identity levels (cellular, connectivity,
                  temporal) - novel framework extension absent
                  from all other table entries.
```

```
key_structural_insight:
  Level of identity failure determines drug class.
  Symptom (tremor) does not determine drug class.
  Same symptom, three levels, three categorically
  different intervention logics.

novel_predictions_locked_2026_03_09:
  1. R_PD (NURR1 / HDAC4/5) predicts Braak stage better
     than TH alone.
  2. GRID2-AAV reduces ET tremor in GRID2-deficient model.
  3. DBS + HDAC4/5i + NURR1 agonist outperforms DBS alone
     (DAT-SPECT endpoint).
  4. Anle138b active against TDP-43 in ALS model
     (cross-entry universal Hub compound prediction).
  5. RBD atonia severity predicts DAT-SPECT decline rate
     (R_RBD as prodromal biomarker of Hub trajectory).
  6. HDAC4/5i + alpha-syn clearance at RBD diagnosis delays
     motor onset by >= 3 years vs. clonazepam alone.
  7. NREM slow-wave amplitude predicts alpha-syn propagation
     rate independently of motor biomarkers (Loop 4).
  8. DBH/LC biomarkers (CSF MHPG) fall before TH/SNc
     biomarkers in prodromal RBD cohort (Braak II before
     Braak III - R_RBD falls before R_PD).
```


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